

Report R19 — the paper figures: sector counts, $D_{\max}$, and how big the sectors actually are

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-12

Abstract

This report accompanies `notebooks/paper_figures.ipynb`, an editable rebuild of R2’s Figure 1 — the number of Krylov sectors against N at `obc0` — together with the matching $D_{\max}$ figure and a new one: the *distribution* of sector sizes for the two exponentially fragmented rules W156 and W108. Three things are worth recording beyond the pictures. First, every curve in the two scaling figures is an **exact integer law**, checked at every computed N rather than fitted, so the annotations the figures carry (F_{N+2} , 2^{N-1} , $\lfloor (N+1)/2 \rfloor + 1$, ...) are certificates; they are listed in Table 1 and guarded by the test suite. Second, W105 and W150 — routinely quoted as one entry “150/105” — differ at $N \equiv 1 \pmod{4}$ and at no other N , in *both* series at once, one extra level of the domain-wall number being exactly one fewer sector; W60 and W102 by contrast agree term for term. Third, the size distribution separates the two exponential rules in a way neither scaling figure does: W156’s is **unimodal**, peaked well away from 1, while W108’s is **monotone decreasing** with 33 552 singletons out of 170 625 sectors at $N = 21$. In both, the sector a random computational-basis state lands in is far smaller than $D_{\max}$ — a factor 4 for W156 and a factor 261 for W108 at $N = 21$ — so $D_{\max}$ alone is a poor description of the typical sector, and the gap widens with N .

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1 What this report is

R2 introduced the two scaling figures and R9 the growth-base plane built on them. What was missing was a version of the figures that can be *edited*: the package figure code in `scaling/figure.py` draws all 256 rules on a fixed template, which is the right tool for a sweep and the wrong one for a paper plate. `notebooks/paper_figures.ipynb` is the replacement. All the drawing code is inline in the notebook, all the adjustable parameters are collected in one cell at the top, and no cell calls the package’s figure module.

What the notebook does *not* own is the arithmetic. The closed forms annotated on the curves live in `scaling/paper_figures.py`, and the notebook imports and re-verifies them before it draws anything. That split is deliberate: a law is a fact about the data, an axis limit is a preference, and only the second should be casually editable.

Outputs. `figures/paper/:` `fig_paper_sectors_obc0.pdf`, `fig_paper_dmax_obc0.pdf`, `fig_paper_scaling_obc0.pdf` (the two as panels (a) and (b) of one plate), `fig_paper_sizehist_obc0.pdf`. Each with a 300 dpi PNG beside it.

2 Every curve is an exact law

rule	tuple	series	closed form	N range	holds
60	IIVV	$D_{\max}$	2^{N-1}	6–21	yes
102	IVIV	$D_{\max}$	2^{N-1}	6–21	yes
105	VIIIV	$D_{\max}$	$\max_w \binom{N+1}{w}$, odd w at $N \equiv 1 \pmod{4}$	6–21	yes
108	IIIV	$D_{\max}$	F_N	6–21	yes
150	IVVI	$D_{\max}$	$\max_{w \text{ even}} \binom{N+1}{w}$	6–22	yes
201	VIII	$D_{\max}$	F_{N+2}	6–21	yes
60	IIVV	n_{wcc}	$N + 1$	6–21	yes
102	IVIV	n_{wcc}	$N + 1$	6–21	yes
105	VIIIV	n_{wcc}	$\lfloor N/2 \rfloor + 1 + [N \equiv 3 \pmod{4}]$	6–21	yes
150	IVVI	n_{wcc}	$\lfloor (N+1)/2 \rfloor + 1$	6–22	yes
156	IIVI	n_{wcc}	F_{N+2}	6–21	yes

Table 1: The exact laws annotated on Figures 2 and 3 (obc0). Each was read off the integer series and then evaluated at *every* computed N ; the last column is that check, not a goodness of fit. F_k is the Fibonacci number with $F_1 = F_2 = 1$.

None of the entries in Table 1 is a regression. Each was read off the integer series and then evaluated at every N in the sweep — 16 or 17 points — and the last column reports that comparison, not a residual. `scaling/paper_figures.verify` performs it, the notebook asserts on it before drawing, and `tests/test_paper_figures.py` runs it independently of either.

Two claims in the figures are recurrences rather than formulae, and are checked the same way:

- W108’s sector count satisfies $a_N = 2a_{N-1} - a_{N-2} + a_{N-3}$ exactly, whose dominant root is $\rho^2 = 1.754878\dots$, the square of the plastic number. R9 obtained that base by fitting and R18 derived it from the wall grammar; the recurrence is a third and purely arithmetic route to it.
- W201’s sector count is W108’s evaluated one site shorter, $n_{\text{wcc}}^{201}(N) = n_{\text{wcc}}^{108}(N - 1)$, at every overlapping N . The two are spin-flip partners, which is why both curves carry the same ρ^{2N} tag and run parallel a constant factor apart.

The one asymptotic statement. W156’s $D_{\max}$ tag reads $4^{N/5}$ and is deliberately *not* in Table 1. At finite N the largest sector is a mixture of rooms of length 2 and 3 (R2 sec. 3), so no exact recurrence holds over the computed range. What is reported instead is the five-step ratio, which needs no fit: $(D_{\max}(21)/D_{\max}(16))^{1/5} = 1.3195079$ against $4^{1/5} = 1.3195079$. Six digits by $N = 21$ is evidence for the derived base; an exact-law assertion would be the wrong claim to make.

2.1 150/105 is not one entry, and 60/102 is

The two polynomial tracks are usually written as pairs, but they are not pairs in the same sense.

W60 and W102 agree *term for term* in both series, so the figure labels that track with an equals sign.

W105 and W150 do not. At $N \equiv 1 \pmod{4}$ — and, over the whole computed range, at no other N — W105’s largest sector carries **odd** domain-wall number where W150’s carries even ($N = 9$: $\binom{10}{5} = 252$ against $\binom{10}{4} = 210$), and at exactly those N W105 has one sector fewer. It is one phenomenon appearing in both panels: an extra accessible level of the wall number is precisely one fewer sector. The figure therefore labels that track with a comma and draws it as two curves — the second rule as a larger open marker underneath the first, so that where the two coincide the reader sees a filled mark inside a ring and where they part sees a lone ring. The four lone rings in the sector panel are at $N = 9, 13, 17, 21$.

3 Figures 1 and 2 — the scaling

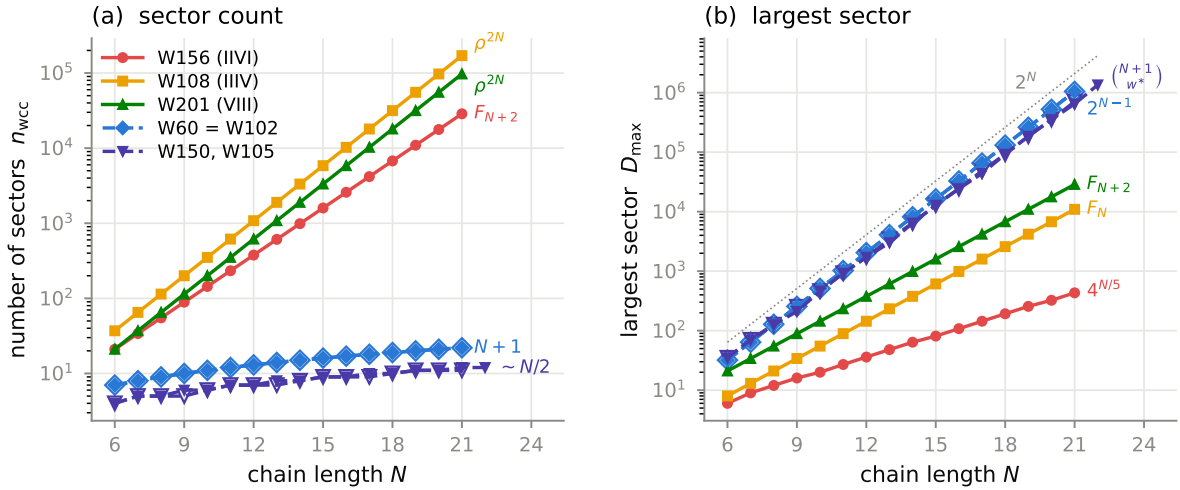


Figure 1: Sector count (a) and largest sector (b) against chain length (obc0, $\log y$). Three rules fragment exponentially — W156 into F_{N+2} sectors, W108 and W201 into $\Theta(\rho^{2N})$ — while W60/W102 and W150/W105 produce only $\Theta(N)$ sectors and correspondingly huge ones: W60’s largest sector is exactly 2^{N-1} , half the entire Hilbert space, and the dotted line is the 2^N ceiling it runs parallel to. Every tag is an exact law (Table 1) except W156’s $4^{N/5}$, which is the derived asymptote. Panels (a) and (b) are near mirror images because $n_{\text{wcc}} \cdot D_{\text{max}} \geq 2^N$.

The two panels are also available separately as `fig_paper_sectors_obc0.pdf` (Figure 2) and `fig_paper_dmax_obc0.pdf` (Figure 3), at single-column proportions, for a paper that wants them apart.

On the drawing. The y axis is a true logarithmic axis with decade ticks rather than $\ln(\cdot)$ on a linear axis, as in the original R2 figures. A straight line still means an exponential, and the reader additionally gets to read the actual sector counts off the axis. The five hues are the CVD-validated categorical order used across the reports, re-checked over all pairs; the worst separation is $\Delta E = 7.2$ (W201 green against W156 red, protanopia), which is admissible only with a secondary encoding, and three are present — distinct markers, a solid/dashed split by family, and a direct law tag at every line end.

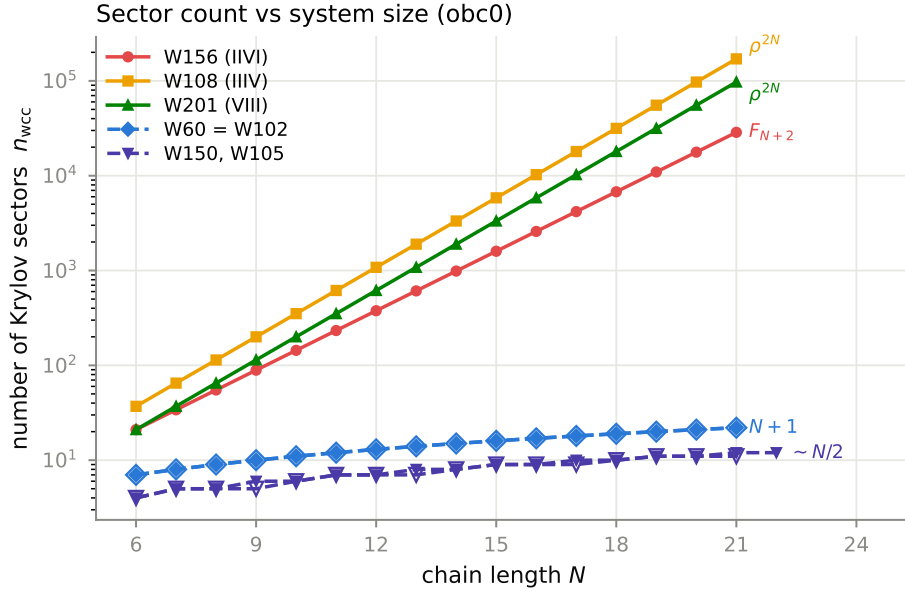


Figure 2: Number of Krylov sectors vs N (obc0), log scale — the R2 Figure 1 content, rebuilt. Exponential fragmentation (156, 108, 201) against the linear and central-binomial families (60/102, 150/105).

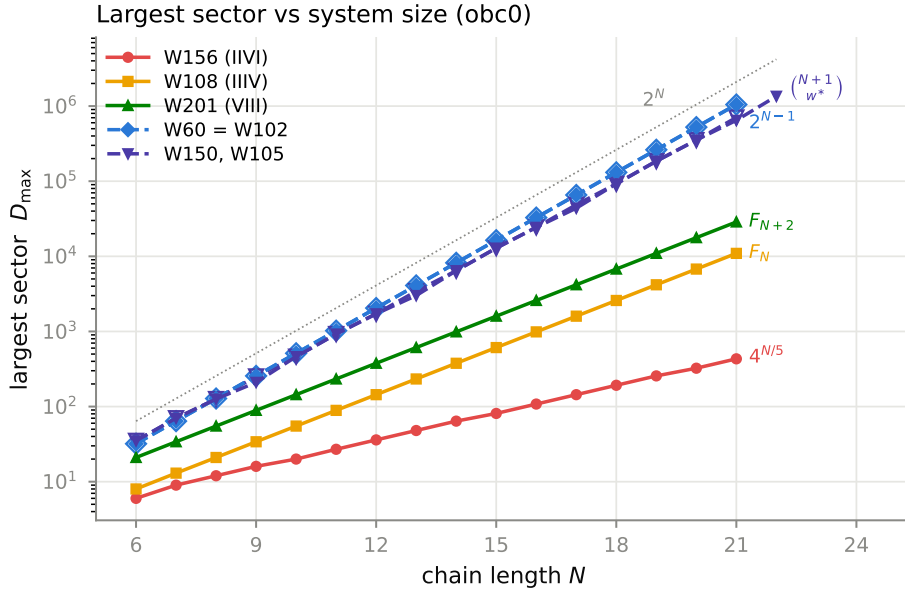


Figure 3: Largest sector $D_{\max}$ vs N (obc0), log scale. The dotted line is 2^N . The ordering is the reverse of Figure 2: the rules with the fewest sectors have the biggest ones.

4 Figure 3 — how big are the sectors?

Neither scaling figure says anything about the *distribution* of sector sizes: n_{wcc} counts them and D_{max} reports one extreme. Figure 4 fills that in for the two exponentially fragmented rules, and the two columns turn out to look nothing alike.

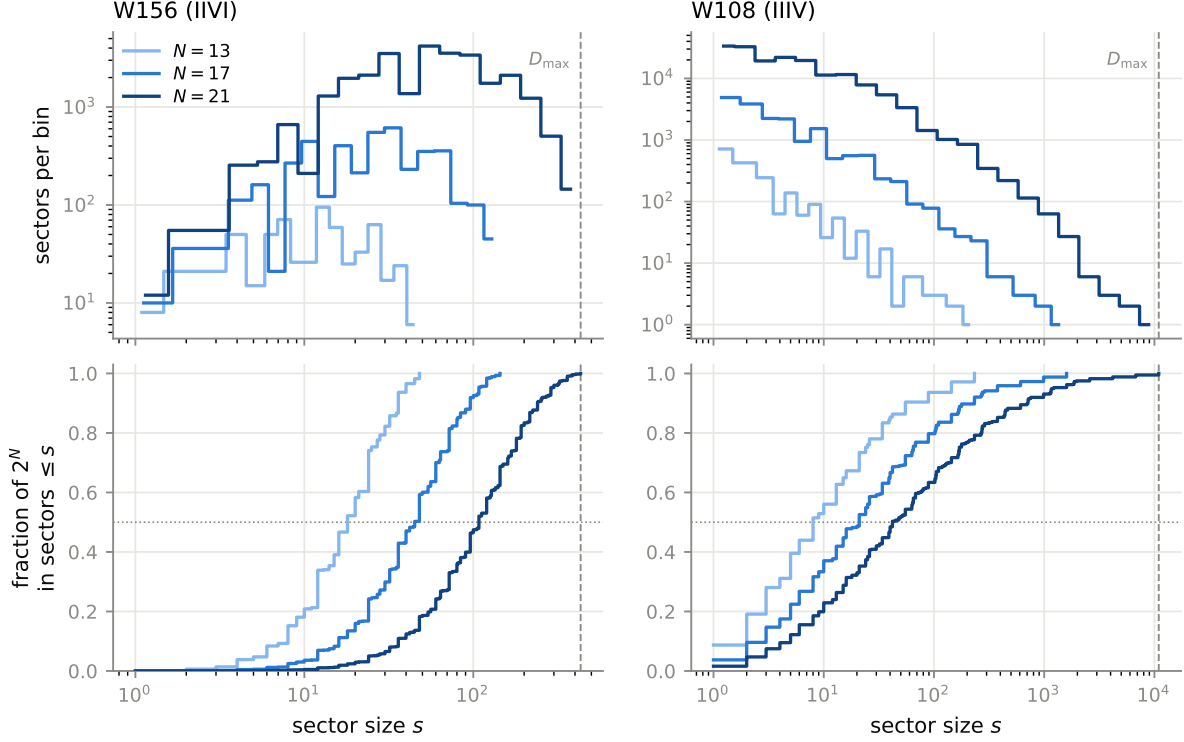


Figure 4: Sector-size distribution of W156 (left) and W108 (right) at $N = 13, 17, 21$ (obc0), light to dark. **Top:** number of sectors per log-spaced size bin. W156 is unimodal and peaked well away from $s = 1$; W108 is monotone decreasing over four decades. **Bottom:** the fraction of the 2^N computational-basis states lying in sectors of size $\leq s$ — read the median sector size off where a curve crosses the dotted line at $1/2$. The dashed vertical rule is D_{max} at $N = 21$; in both rules the typical sector is far to the left of it. The sawtooth in the top-left panel is real and not noise: at small s a bin holds one or two integer sizes, and even sizes are much commoner than odd.

Two different kinds of exponential fragmentation. W156 and W108 sit close together in the growth-base plane — both exponential in both coordinates — and Figure 4 shows that this hides a qualitative difference. W156’s sectors are all of comparable size: at $N = 21$ its 28 657 sectors have mean 73 and median 108, and only 12 of them are singletons. W108’s are not: the same N gives 170 625 sectors of which 33 552, nearly a fifth, are frozen single states, the distribution falls monotonically, and the median is 42 against a D_{max} of 10 946.

D_{max} overstates the typical sector, increasingly. The ratio $D_{\text{max}}/\text{median}$ runs $2.7 \rightarrow 3.2 \rightarrow 4.0$ for W156 over $N = 13, 17, 21$ and $29 \rightarrow 76 \rightarrow 261$ for W108. Both grow, so the gap is not a small- N artefact. This matters wherever D_{max} stands in for “how much of the Hilbert space is dynamically connected”: for W108 at $N = 21$ the largest sector holds 0.5% of the space, but a state drawn uniformly at random sits in a sector of about 42 states out of two million.

rule	N	sectors	$D_{\max}$	$D_{\max}/2^N$	mean	median	singletons
156	13	610	48	5.86e-03	13.4	18	8
156	17	4 181	144	1.10e-03	31.3	45	10
156	21	28 657	432	2.06e-04	73.2	108	12
108	13	1 897	233	2.84e-02	4.3	8	714
108	17	17 991	1 597	1.22e-02	7.3	21	4 895
108	21	170 625	10 946	5.22e-03	12.3	42	33 552

Table 2: The numbers behind Figure 4 (obc0). *mean* is $2^N/n_{\text{wcc}}$; *median* is the size of the sector a uniformly random computational-basis state lands in, i.e. where the cumulative curve crosses $1/2$. Both are far below $D_{\max}$, which is why the largest sector is a poor description of the typical one.

Why this can be computed at all. The stored `sizes_recurrent` list is truncated at 2048 entries for these rules once $N \geq 16$ (W108: $N \geq 14$), which would cap Figure 4 at $N = 15$, or 13 for W108. The `size_hist` field is *not* truncated — it carries every class — and `size_stats` asserts both that its class count equals n_{wcc} and that $\sum_s s h(s) = 2^N$ before returning, so relying on it is checked rather than assumed. That is what puts $N = 21$ within reach.

5 Using and editing the notebook

Open `notebooks/paper_figures.ipynb` and run all cells; it writes into `figures/paper/` and takes a few seconds. Section 2 of the notebook holds every adjustable parameter — page widths, font sizes, the palette, the `TRACKS` list that decides which rules appear and how they are grouped, the law tags and their nudges, the histogram’s N values and bin count, the output directory. The drawing code reads that cell and nothing else.

To add a rule to the scaling figures, append a dict to `TRACKS`. To change which N the histograms show, edit `HIST_NS`. To change a *law*, edit `scaling/paper_figures.py` instead — and expect the test suite to have an opinion.

6 Caveats

- **obc0 only.** The `pb` sweep is held pending caveats (`NEXT_STEPS.md`); under `pb` at odd N the brick-wall layers stop commuting, so those points would need even N only.
- **These are WCC sectors**, which for the unitary rules shown coincide with the strongly connected components, so the decomposition is exact for the monitored and the unmonitored chain alike. That coincidence is a property of unitarity and does not survive dissipation — see R-T14 on the recurrent-class/WCC distinction.
- **One representative per symmetry partner.** W198 is W156 under reflection and W108/W201 are spin-flip partners; the partner is omitted rather than drawn on top of its twin.
- **Nothing here is new dynamics.** The sweep data is R2’s; what is new is the certification of the laws, the 150/105 distinction, and the size distributions.